

Dr. VISHNU BHUVANACHANDRAN

Senior Research Fellow | Mechanical Engineering | Experimental Research | Emission Control
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PROFESSIONAL SUMMARY

Mechanical Engineer with a PhD from NIT Tiruchirappalli (DAE-IGCAR Government Fellow) and 10+ years of research experience in experimental protocol design, industrial device characterisation, emission control, and occupational health. Proven ability to design and execute structured experimental programmes, develop novel test methods, optimise materials and processes for hazard reduction, and coordinate multidisciplinary research teams on government-funded projects. Author of 20+ peer-reviewed publications (SCI/Scopus) and holder of 4 granted Indian patents. Experienced in research leadership as Principal Investigator, student supervision, and disseminating findings to international scientific audiences. Eligible to work in Ireland; available immediately.

KEY SKILLS

Experimental Research: Design of Experiments (DoE), experimental protocol design and execution, emission and particulate characterisation, statistical analysis, failure analysis, data-driven process optimisation

Instrumentation: Personal air samplers, cascade impactors, AWS F1.2 fume chamber, high-volume sampler, NIOSH 7600/7300 methods, elemental analysis (Cr6+, Cr, Ni, Mn, Fe)

Engineering Design: SolidWorks, CAD/Creo, FEA/ANSYS, MATLAB, AutoCAD, material selection, DFM/DFA, rapid prototyping (3D printing, CNC, laser)

Sustainable Engineering: Green consumable development, hazardous emission reduction, materials substitution, process optimisation for reduced environmental impact

Research Management: Principal Investigator on funded projects (INR 42 Lakhs+), project planning and milestone delivery, team coordination (30+ members), financial reporting, procurement

Documentation & Reporting: Technical reports, research proposals, IP filings, compliance reports for safety oversight panels (NIOSH, AWS, DAE), patent authoring

Knowledge Transfer: Industry-academia collaboration (IGCAR-DAE, ANERT, KDISC, GE), IIW international lectures, peer-reviewed publishing, stakeholder presentations

Data Analytics: Python, R, SQL, Power BI, Minitab, Design Expert, statistical modelling, data visualisation

PROFESSIONAL EXPERIENCE

Data Analyst | *Starilum Limited, Drogheda, Ireland*

June 2025 – Present

- Manage and validate complex datasets using SQL and Excel; develop Power BI dashboards for cross-functional reporting.
- Achieved 80% increase in organisational research capabilities within 5 months by improving reporting infrastructure.

Associate Professor, Research Lead & Principal Investigator | *Marian Engineering College, APJ Abdul Kalam Technological University, Kerala, India*

Sept 2019 – Aug 2024

- Principal Investigator on two government-funded projects totalling INR 40 Lakhs (~EUR 43,000): defined project scope, coordinated specialist teams, managed procurement and budget tracking, and delivered structured progress reports to funding bodies.
- Led EV Battery Thermoelectric Cooling System project (ANERT, INR 12 Lakhs): designed system from concept, selected materials, built and tested prototypes — resulting in a granted patent and Best Paper Award.
- Coordinated a 30+ member interdisciplinary team on the Electric Go-Kart project; managed scheduling, quality inspections, and programme milestones — achieving national-level recognition.
- Founded and managed Energy Research Lab and Rapid Prototyping Lab (3D printing, CNC, laser); managed capital equipment and lab governance.
- Served as Assistant Nodal Officer of IEDC and Convener of Institute Innovation Council (IIC); oversaw IP documentation and startup incubations.
- Supervised UG and PG dissertation students; delivered lectures in Mechanical Engineering, Thermal Sciences, and Engineering Data Analysis.
- Published 20+ peer-reviewed papers (SCI/Scopus) and secured 5 Indian patents.

Senior Research Fellow — Environmental Safety & Occupational Health | IGCAR, Dept. of Atomic Energy, Govt. of India | NIT Tiruchirappalli
May 2014 – Jan 2019

- Designed and executed experimental programmes for characterisation and control of welding fumes and sub-micron particulates (PM 2.5, PM1, PM 0.5, PM 0.25) under ISO 15011-1:2009, 10882-1, and AWS F1.2 standards.
- Developed novel nano-composite flux formulations for stainless steel SMAW electrodes, achieving a 70% reduction in carcinogenic Cr(VI) emissions through rigorous DoE-based experimental design — validated through independent replication and resulting in two granted patents.
- Operated specialist sampling equipment: personal air samplers, cascade impactors, high-volume samplers, AWS F1.2 fume chambers; performed elemental analysis using NIOSH 7600 and NIOSH 7300 methods.
- Produced structured compliance and technical reports for government and international safety oversight panels (NIOSH, AWS, DAE); maintained full documentation of methods, results, and corrective actions.
- Coordinated across engineering, chemistry, and safety departments; achieved 90% improvement in process efficiency on sustainability-focused work.
- Delivered two invited lectures at IIW General Assemblies (50+ countries) on welding emission control and green manufacturing.
- Delivered two large-scale Govt. of India strategic R&D projects end-to-end; authored 4 granted patents and 15+ peer-reviewed publications during this period.

Lecturer / Assistant Professor | St. Thomas Institute for Science and Technology, Trivandrum, Kerala
2010–2011 & 2013–2014

- Delivered undergraduate Mechanical Engineering courses; coordinated student placements and skill-based training programmes.

FUNDED RESEARCH PROJECTS

Thermoelectric EV Battery Cooling System | ANERT, Govt. of Kerala | INR 12 Lakhs | Principal Investigator

- Designed and built a dual-stage thermoelectric cooling system for EV battery packs: system layout, material selection, CAD modelling, fabrication, and performance testing. Granted patent No. 202441097925A; Best Paper Award ICNN 2023.

Welding Fume & Particulate Analysis for Nuclear Fabrication | DAE-IGCAR | Senior Research Fellow

- Project 1: Analysis of welding fumes for consumables used in fabrication of nuclear components.
- Project 2: Measurement of PM 2.5, PM1, PM 0.5, PM 0.25 of SMAW fumes for nuclear consumables. Both produced audit-ready documentation and 2 granted patents.

MSME Handloom & Coir Cluster Development | KDISC, Govt. of Kerala | INR 28 Lakhs | Faculty Coordinator

- Coordinated stakeholder liaison with government and industry; maintained documentation and delivered structured progress reports.

Centralised Industrial Vacuum Cleaner | GE Onshore Electricals | INR 2 Lakhs | Principal Investigator

- Industry consultancy: designed system, produced drawings, BoM, and coordinated suppliers through to delivery.

EDUCATION

PhD in Mechanical Engineering — Environmental Safety & Occupational Health 2014 – 2019

National Institute of Technology Tiruchirappalli (NIT-T), India | Advisor: Dr. S. P. Sivapirakasam

Thesis: Investigation on Reduction of Welding Fumes and Cr6+ Formation Using Nano Particle Addition in Stainless Steel SMAW Electrode

Funded by: DAE-IGCAR Senior Research Fellowship, Dept. of Atomic Energy, Govt. of India

MSc in Data Analytics — First Class Honours (President's Gold Medal)

2024 – 2025

Dundalk Institute of Technology, Ireland

M.Tech in Thermal Science — First Class

2011 – 2013

University of Kerala, India

B.Tech in Mechanical Engineering — First Class

2005 – 2009

University of Kerala, India

PEER-REVIEWED PUBLICATIONS (20 PUBLISHED + 1 UNDER REVIEW)

SCI/Scopus-indexed journals in emission control, occupational health, materials engineering, thermal systems, and sustainable manufacturing.

Under Review

1. Akshay B, Faisal Seyas, Vishnu BR, Manu Mohan. Graphene-Silver Hybrid Nanoparticles Embedded Phase Change Materials for Enhanced Thermal Management of Lithium-Ion Batteries. *Journal of Energy Storage*. [Under Review]
2. Yadav S, Sharma S, Sharma M, Agrawal AV, Saraswat A, Williams D, McDonnell J, Wanigasekara J, Kanyaru J, Raval JB, Singh KG, Perera KY, Njeru M, Rushali R, Murnaghan S, Antwi SH, Banchariya S, Singh V, Vishnu BR, Sharma V, Kaushik A. Opinion Paper on the Role of AI in the Advancement of Sustainable Development Goals. [Under Review]
3. Vishnu BR, Kaushik A. Autonomous Health and Fertility Monitoring in Dairy Cattle: Advanced Machine Learning with Calibrated Probabilities and Optimized Decision Thresholds. Elsevier Book Chapter. [Under Review]

Published

4. Rahul M, Sivapirakasam SP, Mohan S, Vishnu BR, Prasanth C. Development of low-emission next-generation stainless steel E304-16 and E316-16 synthetic electrodes with mild steel core wire. *Sustainable Materials and Technologies*. 2025;44:e01372.
5. Vishnu BR, Sivapirakasam SP, Mohan S, Rahul M. Investigations on the weld metal properties of nanoparticle-embedded low-fume SMAW electrodes. *Transactions of the Indian Institute of Metals*. 2023;76(3):675-84.
6. Rahul M, Sivapirakasam SP, Sreejith M, Vishnu BR et al. Concurrent Reduction of Fume and Cr(VI) Concentrations Using a New Covered SS SMAW Electrode. *Journal of Environmental Chemical Engineering*. 2024. DOI:10.1016/j.jece.2023.111632
7. Kannan RR, Sivapirakasam SP, Rahul M, Mohan S, Vishnu BR. Effect of Activated Fluxes on BZC and Cr(VI) Concentration in Stellite Hardfacing. *Materials Today Communications*. 2023.
8. Rahul M, Paramashivan SS, Mohan S, Vishnu BR. Study on soluble and insoluble fume and Cr(VI) emitted from a new covered electrode with nano-titanate flux. *Environmental Science and Pollution Research*. 2023;30(42):95550-95565.
9. Vishnu BR, Paramashivan SS, Mohan S, Albert SK. Effect of substituting fine rutile with nano TiO₂ on mass transfer efficiency and welding fume reduction in SS SMAW electrode. *High Temperature Materials and Processes*. 2020;39(1):117-23.
10. Vishnu BR, Sivapirakasam SP, Satpathy KK, Albert SK, Chakraborty G. Influence of nano-sized flux materials in reduction of Cr(VI) in SS welding fumes. *Journal of Manufacturing Processes*. 2018;34:713-20.
11. Vishnu BR, Sivapirakasam SP, Satpathy KK, Albert SK, Chakraborty G. Cr⁶⁺ reduction in welding fumes by nano composite coatings on SS MMAW electrodes. *Process Safety and Environmental Protection*. 2018;114:334-46.
12. Kannan RR et al., Vishnu BR. Reduction of Cr⁶⁺ in stellite hardfacing by zinc coating on filler rod. *Welding in the World*. 2023;67(1):21-31.
13. Rahul M et al., Vishnu BR. Study on mass concentration and morphology of SMAW fume particles with nano-CaTiO₃ arc stabilizer. *Journal of Manufacturing Processes*. 2022;84:230-9.
14. Madhusoodhanan R et al., Vishnu BR. Manufacture of low fume welding electrode using synthetic rutile flux. *International Journal of Advanced Manufacturing Technology*. 2022;121(11-12):8197-208.
15. Kannan RR et al., Vishnu BR. Reduction of Cr⁶⁺ in welding fumes during stellite hardfacing with GTAW. *Materials Today: Proceedings*. 2020;27:2052-5.
16. Mohan S et al., Vishnu BR. Corrosion behaviour of ZrO₂-TiO₂ nano composite coating on SS under simulated marine environment. *Materials Today: Proceedings*. 2020;27:2492-7.
17. Kannan RR et al., Vishnu BR. Novel activated flux coating method for reducing fumes at BZC for stellite hardfacing. *Materials Today: Proceedings*. 2021;46:9246-51.
18. Rahul M et al., Vishnu BR. Health issues owing to welding fumes and control strategies — A review. *Materials Today: Proceedings*. 2021;46:9239-45.
19. Rahul M et al., Vishnu BR. Experimental investigation on gripper force of electrically activated non-contact swirl vane gripper. *Materials Today: Proceedings*. 2021;46:9636-40.
20. Mohan S et al., Vishnu BR. Performance and emission analysis of an IC engine with hydroxy gas assist. *Materials Today: Proceedings*. 2021;46:10043-6.
21. Rao MJ et al., Vishnu BR. Prediction and analysis of semi low floor bus driver seat vibrations by response surface methodology. *International Journal of Heavy Vehicle Systems*. 2021;28(3):409-22.
22. Joseph A et al., Vishnu BR. Pool boiling heat transfer enhancement using sol-gel derived nano-CuO porous coating. *Experimental Thermal and Fluid Science*. 2019;103:37-50.
23. Mohan S et al., Vishnu BR. Performance Evaluation of Linear Solar Collector Using Hybrid Nanofluid. In: I-DAD 2020. Springer Singapore, 2021, pp. 465-472. [Book Chapter]

BOOK EDITING

Vishnu BR, Kaushik A, Sharma S (Editors). Artificial Intelligence for Sustainable Futures: Innovations in Data Science, Analytics, and Machine Learning towards Global Impact. Taylor & Francis. [In Progress]

Note: Editing a book for Taylor & Francis as lead editor, covering AI applications in data science, machine learning, and sustainable development — a significant independent academic achievement reflecting international research leadership.

PATENTS — 4 GRANTED + 1 PUBLISHED (AWAITING GRANT)

- Indian Patent No. 201741039032 — A Novel Flux Composition for Stainless Steel SMAW Electrodes. [Granted]
- Indian Patent No. 201741042166 — An Industrial Helmet for Welding Fume Expulsion. [Granted]
- Indian Patent No. 202041025422 — Thermal Spray Coating Attachment for Cylindrical Substrate. [Granted]
- Indian Patent No. 202141005994 — A Novel Flux Composition for Stainless Steel SMAW Electrodes with Mild Steel Core Wire. [Granted]
- Indian Patent No. 202441097925A — Thermoelectric Enhanced Liquid Dual Stage Cooling Solution for Swappable EV Battery Packs. [Published; Grant Awaiting]

INTERNATIONAL PRESENTATIONS

- Invited Lecture: 'Source Reduction Strategies for Arc Welding Fumes — Way Forward to Green Welding Process.' IIW 2021 General Assembly, July 14, 2021.
- Invited Lecture: 'Investigations on Control Strategies for Welding Fumes and Cr(VI) Formation Rate from SS SMAW by Nano-Particle Additions.' IIW Commission VIII — Health Safety & Environment UK (VIII-2304-21), March 16, 2021.
- 'Nanoparticles Impregnated PCM for Augmented Thermal Management in Battery Modules.' ICNN 2023, VNIT Nagpur, India.
- 'A Characteristic Study on Welding Fumes From MMAW Electrodes.' National Welding Seminar, IIW India, Jamshedpur, January 2015.

CERTIFICATIONS & TRAINING

- ISO 13485 — Quality Management System for Medical Devices (Udemy)
- PLM Fundamentals — BoM management, revisions, ECR/ECO workflows (Udemy)
- ISO 15011-1:2009, 10882-1, AWS F1.2 — Welding Emission Characterisation & Test Setup
- Advanced Microsoft Power BI | Advanced MS Office Suite (Skill Nation)

AWARDS & RECOGNITION

- President's Gold Medal — MSc Data Analytics, First Class Honours, Dundalk Institute of Technology (2025)
- Best Paper Award — Dual Stage Thermoelectric Cooling Method for EV Batteries (ICNN 2023, VNIT Nagpur)
- DAE-IGCAR Senior Research Fellowship — Dept. of Atomic Energy, Govt. of India
- National Award — Electric Go-Kart Innovation Project
- Invited Speaker — IIW International General Assemblies (2021)

REFERENCES

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